

Notebook-Evo · Project Showcase

AI Knowledge Workspace · Privacy-first · Multi-model · Self-hostable

For AI / Full-stack engineering roles | 2026-08

Project Summary

A deep enhancement of **Open Notebook** (MIT). Users upload research materials, then ask questions against a **hybrid retrieval + knowledge graph + GraphRAG** stack, generate reports, and orchestrate multi-agent workflows. The system supports **phone / desktop / tablet** real-time sync (WebSocket + SurrealDB LIVE query), is fully local, and has zero external service dependency.

Tech Stack

Backend	Python 3.12 · FastAPI · LangGraph · SurrealDB · Esperanto
Frontend	Next.js 16 · React 19 · Tailwind · shadcn/ui · 14 langs i18n
Retrieval	BM25 (jieba for CJK) · Vector · RRF · Rerank · HyDE
AI Orchestration	LangGraph state · Multi-agent persistence · MCP Server
Deployment	Docker Compose · Windows one-click · Daily auto-backup
Testing	360+ pytest · Frontend lint · CI lint+test

Key Capabilities (12 major enhancements)

- Hybrid retrieval: BM25 (jieba + rank_bm25 fallback for CJK) + vector recall + RRF fusion + Rerank
- Knowledge graph + GraphRAG: per-notebook entity/relation extraction, visualization, graph-enhanced QA
- RAG evaluation centre: RAGAS four metrics (Faithfulness / Relevancy / Precision / Recall) auto-evaluated
- Multi-agent orchestration: LangGraph state machines persisted across restarts, task DAG dependencies
- MCP Server: knowledge-base capabilities exposed as standard MCP tools, callable from Claude / Cursor
- Cross-provider model fallback: xcode.best → SenseNova → OpenRouter, priority configurable
- Security engineering: SSRF / DNS-rebinding protection, SurrealQL injection hardening, Jinja2 template-injection fix
- Authentication: constant-time password compare (secrets.compare_digest) + Swagger protection + public tunnel access control
- Engineering quality: 360+ unit tests, CI lint+test, Docker one-click deploy, Windows one-click demo scripts
- Operations automation: daily auto-backup (scheduled task), health-check auto-restart, Tailscale private network
- Mobile PWA: localized manifest + PNG icons + drawer-style navigation + share-by-QR for notebooks
- 14-language i18n: zh-CN / en-US / ja-JP / ko-KR / fr-FR / de-DE etc.

Independent bug fixes (selected)

Bug: Source Chat cannot see the source body Root cause: ContextBuilder.build() ignored context_config.sources and always used inclusion_level='insights', so full_text was never sent to the LLM Fix: build() reads cfg_sources + source_chat.py explicitly passes 'full content'	Context grew from 43 tokens to 2305 tokens; source chat now actually reads the document
Bug: Every share link was broken Root cause: 'token' is a SurrealDB protected variable; using it as a query parameter raised 'Protected variable name' Fix: query parameter renamed to \$share_token	ShareLink.get_by_token returns the link, share feature restored
Bug: source_count always 0 / search never filtered by notebook Root cause: source->reference->notebook relations had inverted in/out in every SQL query Fix: notebooks.py and sources.py queries use the correct direction	source_count correct; search filters by notebook; shared notebook renders all sources
Bug: Async commands (embed_note etc.) always 'Command not found' Root cause: under uvicorn reload, the request-handling subprocess had an empty command registry Fix: api/main.py imports commands modules at startup	registered 10 commands; note/source embeddings actually run
Bug: SenseNova API returned 403 for every request Root cause: wrong endpoint (api.sensenova.cn all forbidden) Fix: SENSENOVA_BASE_URL set to https://token.sensenova.cn/v1	deepseek-v4-flash verified working; default chat model resolved

Engineering Highlights

- Git hygiene: clean commit graph (feat / fix / perf / docs labels), README / CHANGELOG / CONTRIBUTING / SECURITY all present
- CI pipeline: lint + 360+ unit tests, all green
- Config-as-docs: .env.example has comments for every key; MODEL_PROVIDER_PRIORITY lets users adjust model fallback by changing one line
- Observability: /api/system/status returns DB connection state, worker running, model count, db_stats in real time
- Developer experience: health-check.ps1 auto-restarts services, demo-mode.ps1 one-click demo, get-tunnel-url.ps1 surfaces the public tunnel URL

Demo and Source

- GitHub: <https://github.com/SCurrya/notebook-evo>
- README + architecture diagram + API docs + deployment guide + CHANGELOG are all in place
- Public demo (Cloudflare tunnel): see the latest URL in the README
- Demo video (phone screen capture): link will be added to README shortly

Beyond Code

- Security hygiene: .env in .gitignore, GitHub Push Protection caught a hardcoded OpenRouter key, all credentials read from .env

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- Engineering discipline: each commit is an independently workable snapshot; local HEAD verified equal to remote HEAD via ls-remote
 - Practical debugging: Stop-Process leaves zombie sockets that cannot be killed; workaround was switching to a different port and recommending a full restart
 - User-first: turning 'I think it works' into 'users actually like it' (drawer mobile nav, share QR codes, persistent search history)